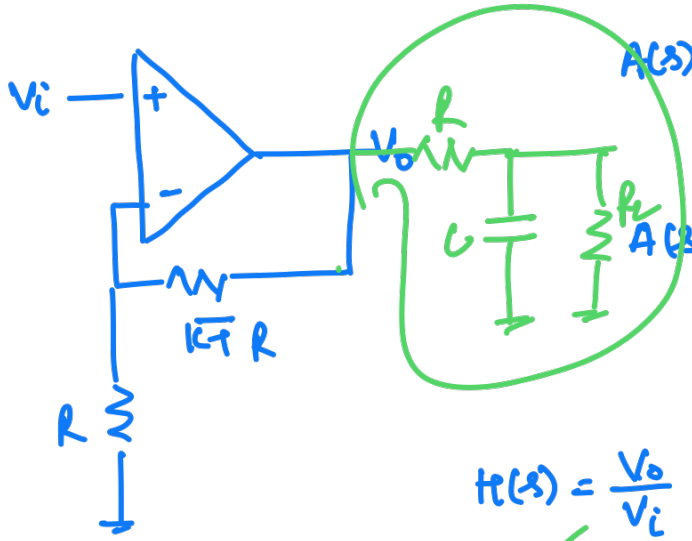
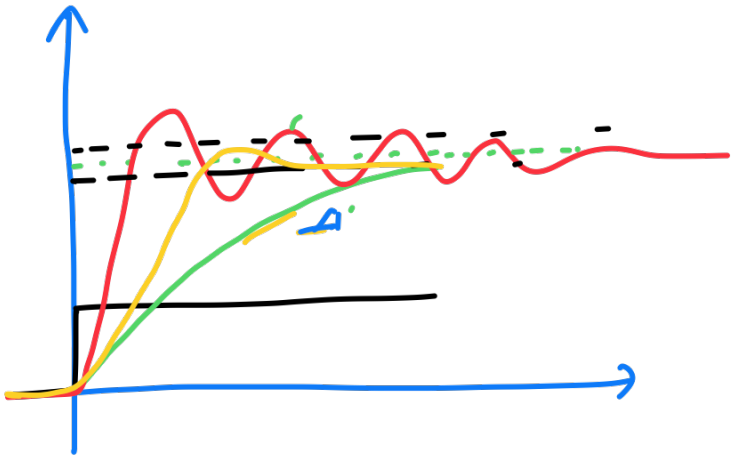
Lecture #13



$$A(s) = \frac{A_{02}}{(1+s/\omega_p)^2}$$

$$A(s) = \frac{A_{02}}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}$$

$$H(s) = \frac{V_o}{V_i} = \frac{K/(1+K/A_{02})}{1+\frac{1}{1+\frac{A_{02}}{K}}\frac{s(\omega_{p1}+\omega_{p2})}{\omega_{p1}\omega_{p2}}+\frac{1}{(1+\frac{A_{02}}{2})\omega_{p1}\omega_{p2}}s^2}$$



$$D(s) = \left[s^2 + s(\omega_{p1}+\omega_{p2}) + \left(1+\frac{A_{02}}{K}\right)\omega_{p1}\omega_{p2}\right] \times \frac{1}{\left(1+\frac{A_{02}}{K}\right)\omega_{p1}\omega_{p2}}$$

$$s_1+s_2 = -\frac{\omega_{p1}+\omega_{p2}}{1}$$

$$s_1 s_2 = \left(1+\frac{A_{02}}{K}\right)\omega_{p1}\omega_{p2}$$

$$s_1 \ll s_2 \Rightarrow s_1+s_2 \approx s_2 = -\left(\frac{\omega_{p1}+\omega_{p2}}{1}\right) = -\omega'_{p2}$$

$$s_1 = -\left(1+\frac{A_{02}}{K}\right)\frac{\omega_{p1}\omega_{p2}}{\omega_{p1}+\omega_{p2}} = -\omega'_{p1}$$

$\omega_{p1} = 10^4$

$\omega_{p2} = 10^7$

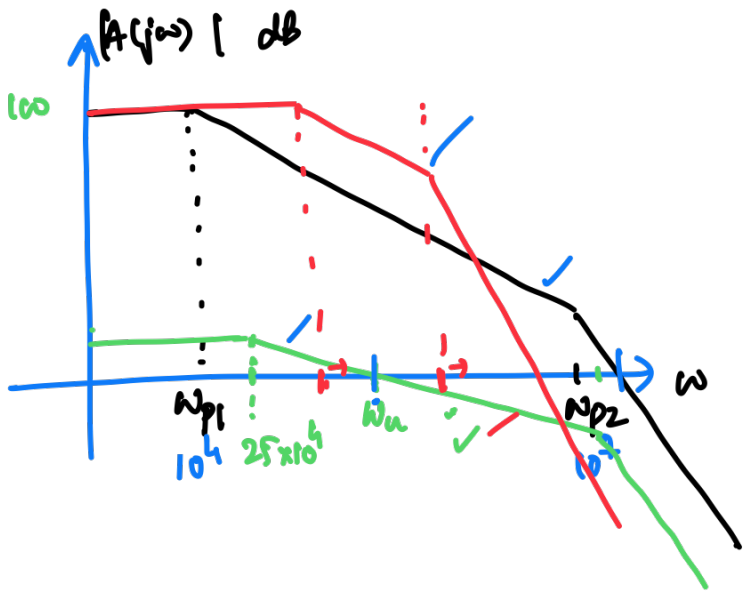
$A_{02} = 100$

$K = 4$

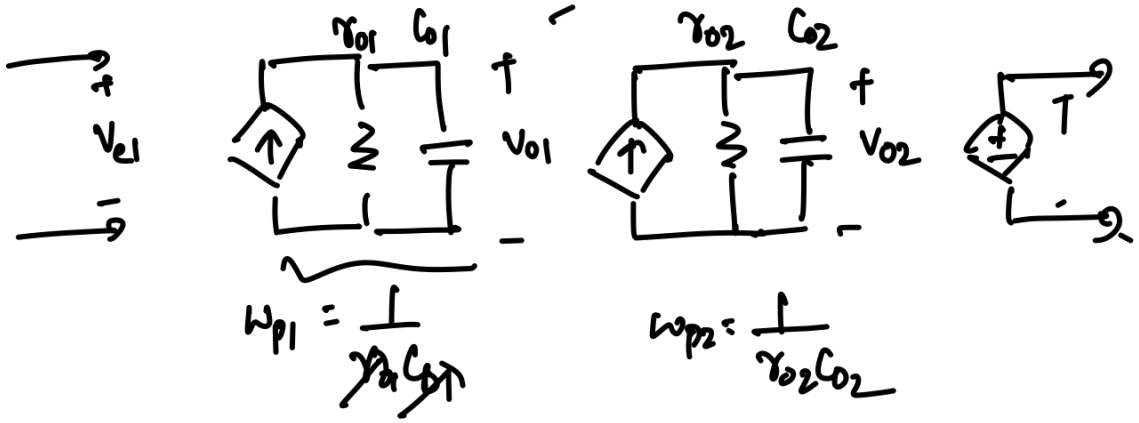
$$\omega'_{p2} = 10^7+10^4$$

$$\omega'_{p1} = \frac{10^7\times10^4}{10^7+10^4}\left(1+\frac{100}{4}\right) = 25\times10^4$$

$$\frac{A_c}{1+A_c/A_{02}} = \frac{A_0}{1+A_0/\omega_0}$$



$$H(s) = \frac{\frac{R_C}{(1+R_C/A_{02})}}{(1+s/\omega'_{p1})}$$



$$\omega_{p1} = \frac{1}{r_{o1}C_{o1}} \qquad \omega_{p2} = \frac{1}{r_{o2}C_{o2}}$$

$$A(s) = \frac{A_{03}}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)\left(1+\frac{s}{\omega_{p3}}\right)} \quad ; \quad \omega_{p1} < \omega_{p2} < \omega_{p3}$$

$$\frac{V_o}{V_i} = H(s) = \frac{R_C}{1 + \frac{R_C\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)\left(1+s/\omega_{p3}\right)}{A_{03}}}$$

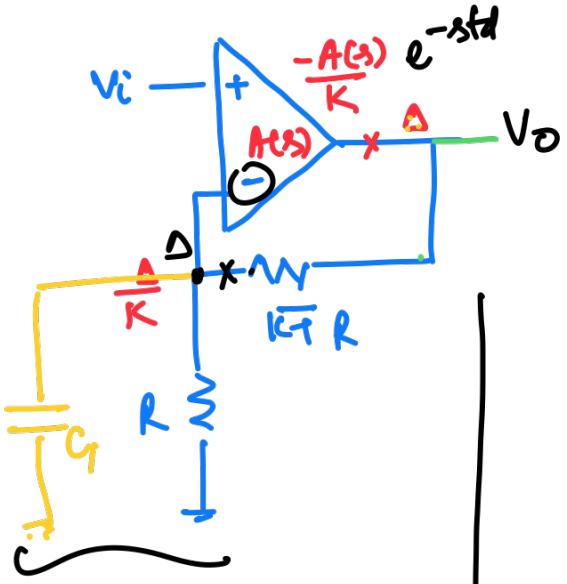
$$D(s) = 1 + \frac{R_C}{A_{03}}\left(1+\frac{s}{\omega_{p1}}\right)\left(1+s\left(\frac{1}{\omega_{p2}}+\frac{1}{\omega_{p3}}\right)+\frac{s}{\omega_{p2}\omega_{p3}}\right)$$

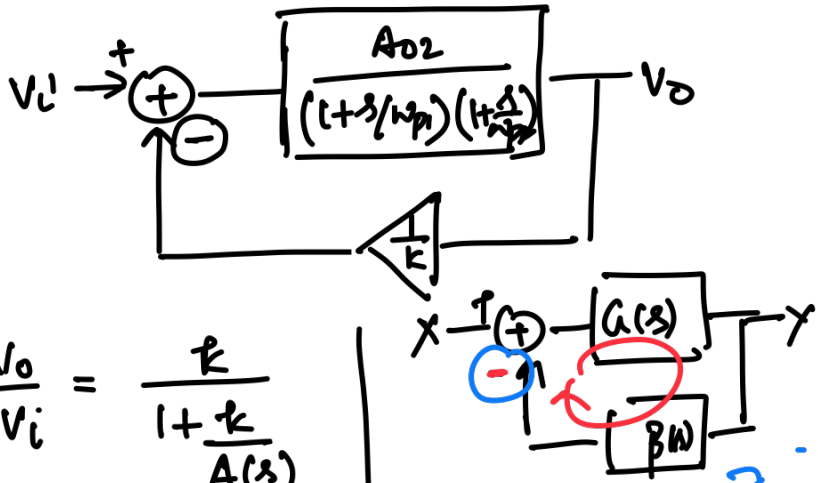
$$= 1 + \frac{R_C}{A_{03}}\left(1+s\left(\frac{1}{\omega_{p1}}+\frac{1}{\omega_{p2}}+\frac{1}{\omega_{p3}}\right)+\frac{s^2}{1}\left(\frac{1}{\omega_{p1}\omega_{p2}}+\frac{1}{\omega_{p1}\omega_{p3}}\right.\right.$$

$$+\frac{1}{\omega_{p2}\omega_{p3}}\Big) + \frac{s^2}{\omega_{p1}\omega_{p2}\omega_{p3}}\Big) -$$

$s \to$

- All poles must be in L.H.P
- Nyquist Stability Criterion → All poles system. "Loop gain & Phase margin"





- Loop gain

i/p: Δ

o/p: $\ominus \frac{A \cdot \Delta}{K}$

$$\frac{V_o}{V_i} = \frac{k}{1+\frac{k}{A(s)}} = \frac{kc}{1+\frac{1}{LG}}$$

$$\frac{V_o}{V_i} = \frac{kc \cdot LG}{(1+LG)}$$

$$\frac{Y}{X} = \frac{G(s)}{1+G(s)\beta(s)} = \frac{G(s)}{1+LG}$$

$$= [X(s) - \beta Y(s)]\, G(s) = Y(s)$$

$LG = G(s)\beta(s)$

Gain around the loop, $LG = \frac{A}{K}$

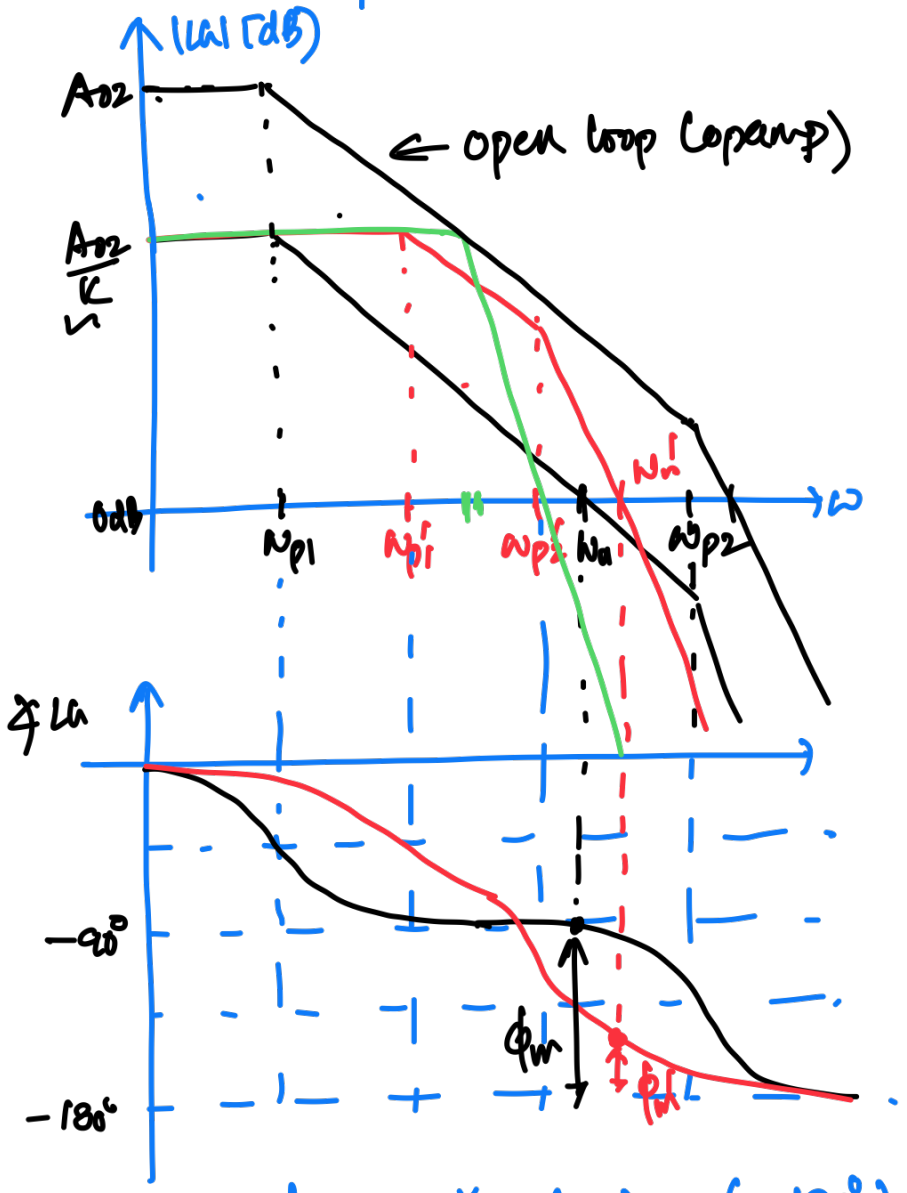
open loop gain

$$LA(s) = \frac{A_{02}}{K\left(1+\frac{s}{\omega_{p1}}\right)\left(1+s/\omega_{p2}\right)}$$

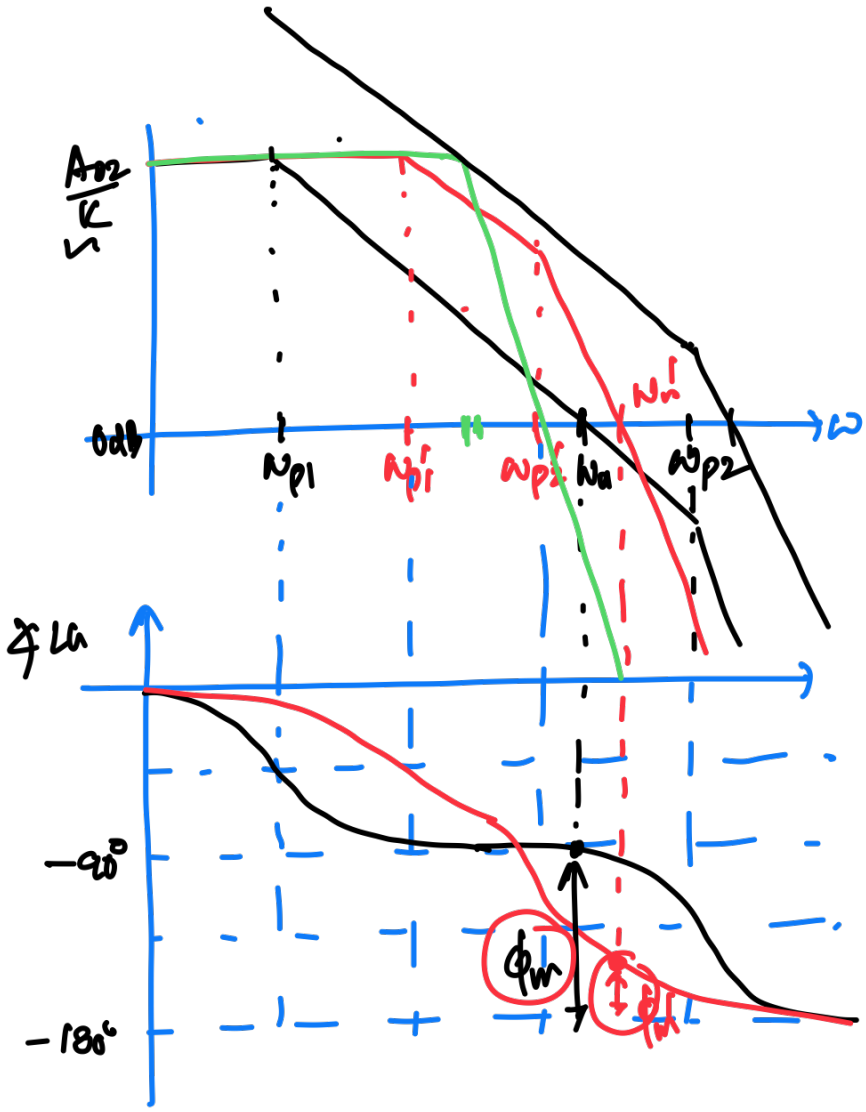
$1 + LA = 0$

$|LA| = 1$

$\angle LA = 180°$.



Phase margin, $\phi_m = \angle LA(\omega_u) - (-180°)$



$\phi_m > 70^\circ$

$\phi_m = 45^\circ$

$\boxed{\phi_m \geq 45^\circ}$